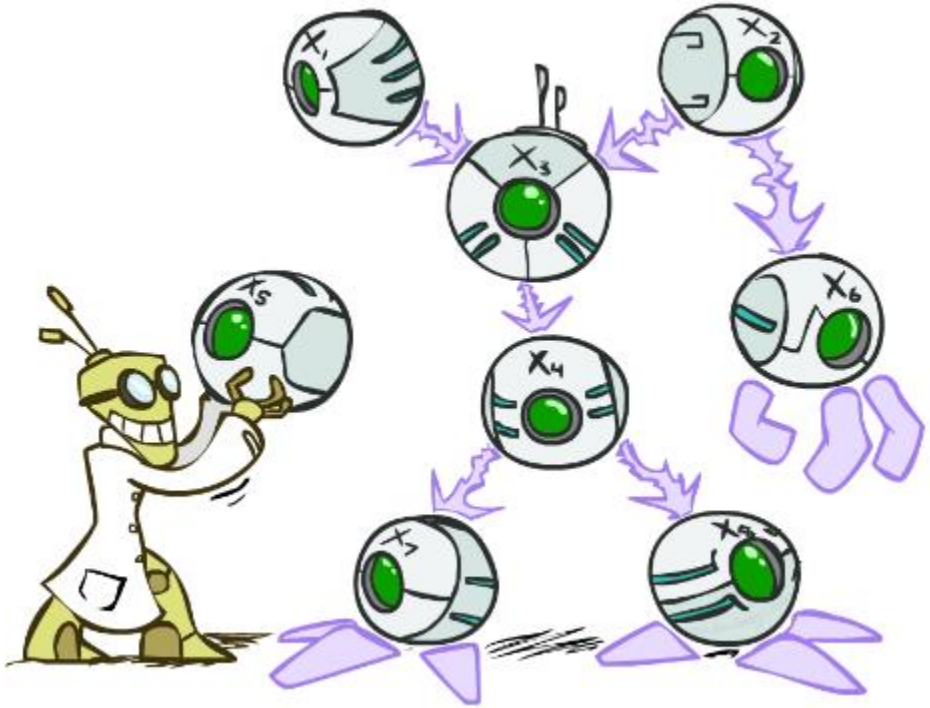
CS 188: Artificial Intelligence

Bayes' Nets – part 2



Instructors: Dan Klein and Pieter Abbeel --- University of California, Berkeley

[These slides were created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All CS188 materials are available at http://ai.berkeley.edu.]

Modified by Toon Calders

Recap: Motivating Example

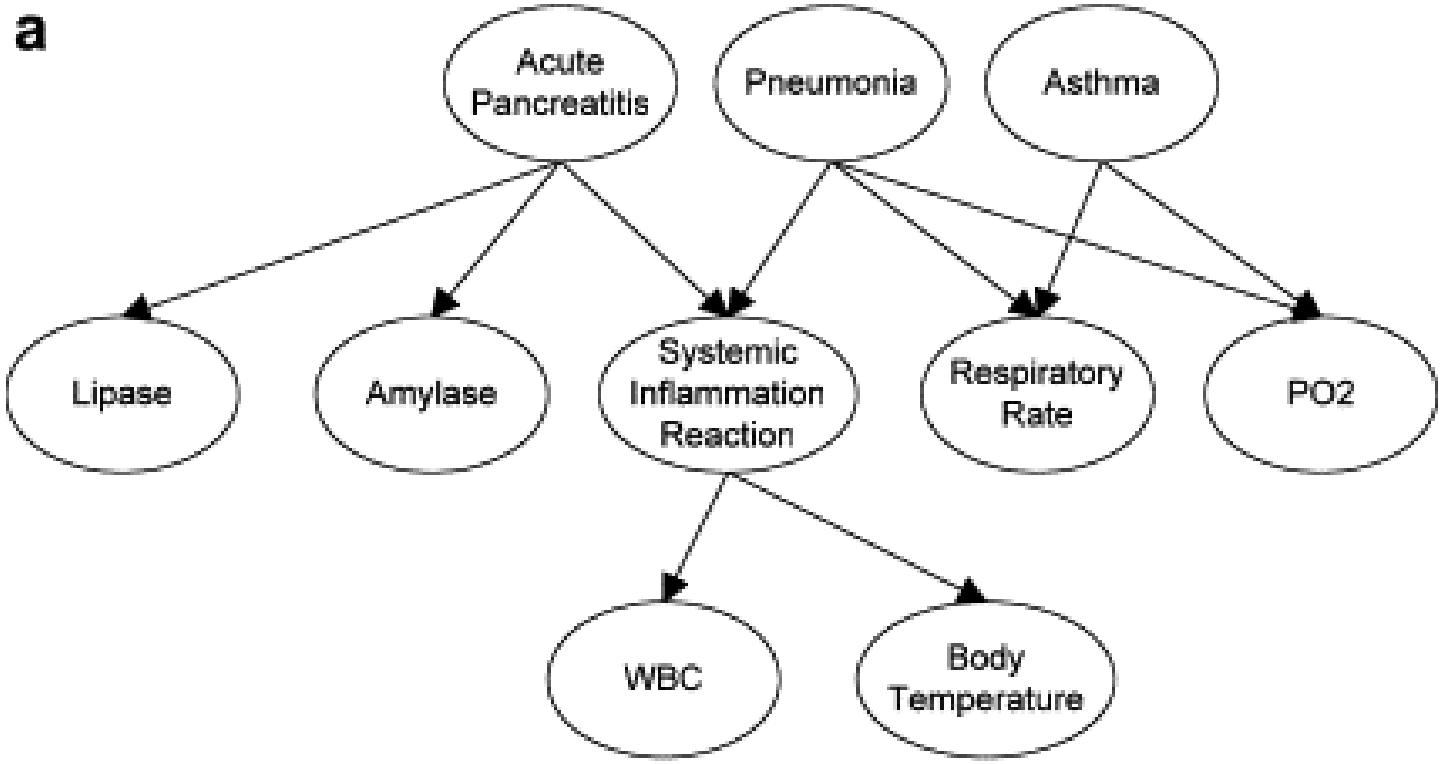
- *Our patient is a known Asthma patient, has an elevated blood cell count, an elevated level of amylase, but a normal body temperature. What is the probability that our patient has a systemic inflammation reaction?*



- How to model our knowledge?
- How to make inferences?

Motivating Example: Bayesian Net

a



b

Pneumonia	Asthma	PO2	Probability
Positive	Positive	<60	0.6
		60~80	0.2
		>80	0.2
Positive	Negative	<60	0.3
		60~80	0.3
		>80	0.4
Negative	Positive	<60	0.4
		60~80	0.3
		>80	0.3
Negative	Negative	<60	0.05
		60~80	0.1
		>80	0.85

Lin, Jau-Huei, and Peter J. Haug. "Exploiting missing clinical data in Bayesian network modeling for predicting medical problems." *Journal of biomedical informatics* 41.1 (2008): 1-14.

Making Inferences

- Given a Bayesian Net, determine P(Q|E=e)
  - Q: outcome or "Query" we are interested in
  - E: "Evidence" we have
  - H: other variables; "Hidden" or unknown

$$
\begin{aligned}
P(Q|E=e) &= \frac{P(Q, E=e)}{P(E=e)} \\
&\propto P(Q, E=e) \\
&= \sum_h P(Q, E=e, H=h)
\end{aligned}
$$

P(E=e) is constant

Can be decomposed using the Bayesian Net structure and computed with the conditional pribability tables

Note on P(Q | E=e) vs P(Q, E=e)

- If we have the distribution P(Q, E=e), we automatically have P(Q|E=e) as well

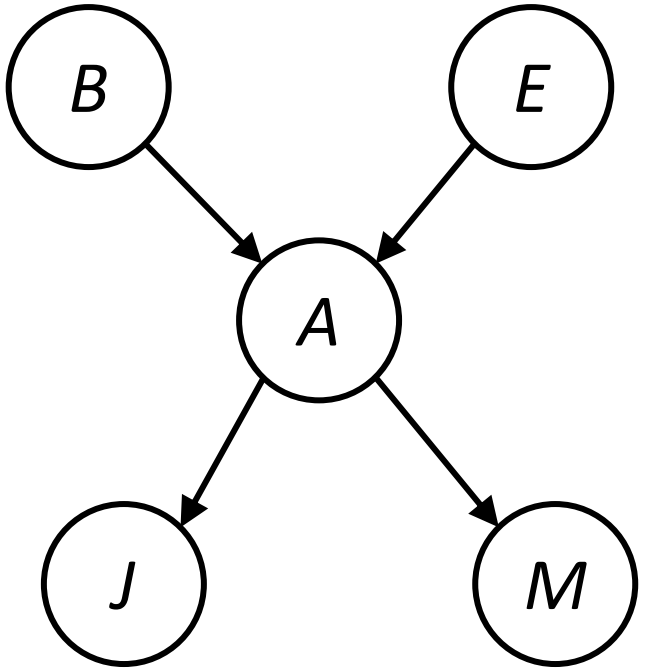
$$P(E = e) = \sum_q P(Q = q, E = e)$$

$$P(Q|E = e) = \frac{P(Q, E = e)}{\sum_q P(Q = q, E = e)}$$

- Hence, we normalize the vector P(Q,E=e)
  - Often Z is used to denote the normalization factor $\sum_q P(Q = q, E = e)$

Inference by Enumeration in Bayes' Net

- Given unlimited time, inference in BNs is easy
- Reminder of inference by enumeration by example:



$$P(B \mid +j, +m) \propto_B P(B, +j, +m)$$

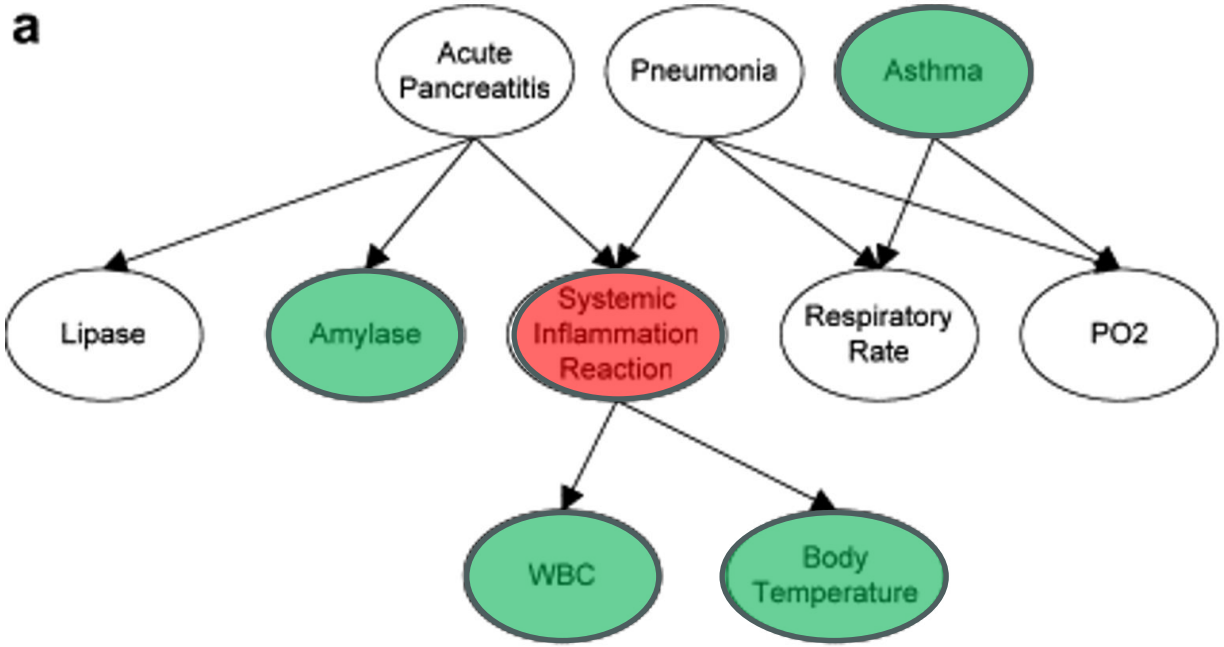
$$= \sum_{e,a} P(B, e, a, +j, +m)$$

$$= \sum_{e,a} P(B)P(e)P(a|B,e)P(+j|a)P(+m|a)$$

$$= P(B)P(+e)P(+a|B,+e)P(+j|+a)P(+m|+a) + P(B)P(+e)P(-a|B,+e)P(+j|-a)P(+m|-a)$$
$$P(B)P(-e)P(+a|B,-e)P(+j|+a)P(+m|+a) + P(B)P(-e)P(-a|B,-e)P(+j|-a)P(+m|-a)$$

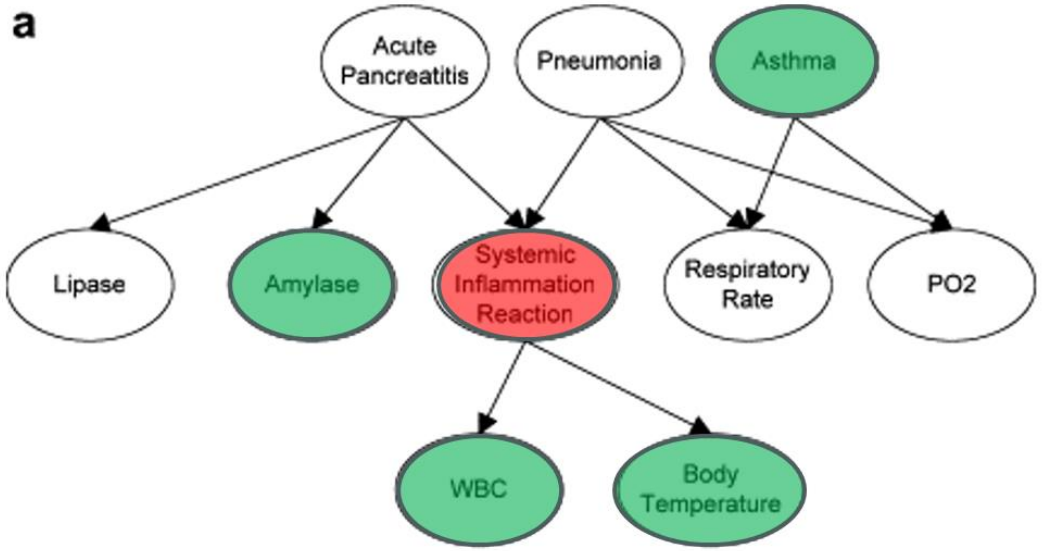
Making inferences

Our patient is a known Asthma patient, has an elevated blood cell count, an elevated level of amylase, but a normal body temperature. What is the probability that our patient has a systemic inflammation reaction?



$$P(S \mid +am, +as, +w, -b)$$

Making inferences



$$P(S|+as,+am,+w,-b)$$

$$= \frac{P(S,+as,+am,+w,-b)}{P(+as,+am,+w,-b)}$$

$$\propto P(S,+as,+am,+w,-b)$$

$$= \sum_{ac}\sum_{pn}\sum_{l}\sum_{r}\sum_{po} P(ac,pn,l,r,po,S,+as,+am,+w,-b)$$

$$= \sum_{ac}\sum_{pn}\sum_{l}\sum_{r}\sum_{po} P(ac)P(pn)P(+as)P(l|ac)P(+am|ac)P(S|ac,pn)$$

$$\times P(r|pn,+as)P(po|pn,+as)P(+w|S)P(-b|S)$$

Making Inferences More Efficient

- Rearranging terms makes inference more efficient

$$\sum_{ac}\sum_{pn}\sum_{l}\sum_{r}\sum_{po} P(ac)P(pn)P(+as)P(l|ac)P(+am|ac)P(S|ac,pn)$$
$$\times P(r|pn,+as)P(po|pn,+as)P(+w|S)P(-b|S)$$

Summing up 2^5 terms of 10 factors each = 31+(32x9) = **319 operations**

$$= P(+w|S)P(-b|S)P(+as)\sum_{ac} P(ac)P(+am|ac)\sum_{pn} P(pn)P(S|ac,pn)$$
$$\times \sum_{r} P(r|pn,+as)\sum_{po} P(po|pn,+as)\sum_{l} P(l|ac)$$

70 operations

We can do even better! How?

Making Inferences More Efficient

- Some variables do not play any role:

$$P(+w|S)P(-b|S)P(+as)\sum_{ac}P(ac)P(+am|ac)\sum_{pn}P(pn)P(S|ac,pn)$$

$$\times\sum_{r}P(r|pn,+as)\sum_{po}P(po|pn,+as)\sum_{l}P(l|ac)$$

$$= \quad P(+w|S)P(-b|S)P(+as)\sum_{ac}P(ac)P(+am|ac)\sum_{pn}P(pn)P(S|ac,pn)$$

10 operations

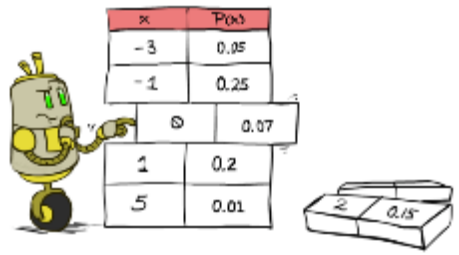
$$\propto \quad P(+w|S)P(-b|S)\sum_{ac}P(ac)P(+am|ac)\sum_{pn}P(pn)P(S|ac,pn)$$

Compute from inside to outside; e.g., avoid looking up P(+pn) multiple times

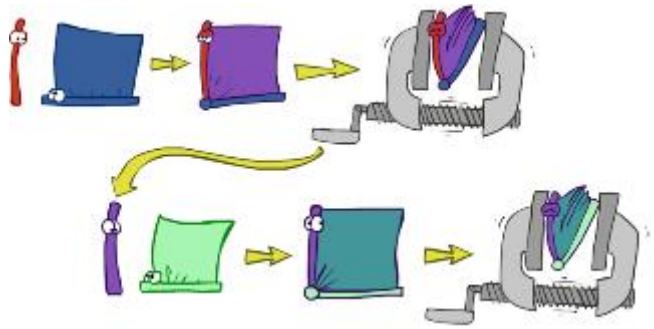
- Hide evidence node A if $Q \perp\!\!\!\perp A \mid E \setminus \{A\}$
- Drop hidden nodes that are no ancester of query or evidence node

General Variable Elimination

- Query: $P(Q|E_1 = e_1, \ldots E_k = e_k)$



- Start with initial factors:
  - Local CPTs (but instantiated by evidence)



- While there are still hidden variables (not Q or evidence):
  - Pick a hidden variable H
  - Join all factors mentioning H
  - Eliminate (sum out) H

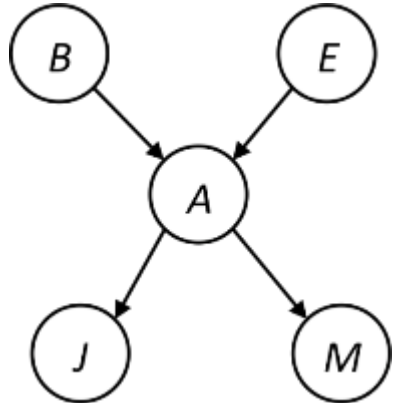
- Join all remaining factors and normalize



$\times \frac{1}{Z}$

Same Example in Equations

$$P(B|j,m) \propto P(B,j,m)$$



$P(B)$	$P(E)$	$P(A\|B,E)$	$P(j\|A)$	$P(m\|A)$

$$\begin{aligned}
P(B|j,m) &\propto P(B,j,m) \\
&= \sum_{e,a} P(B,j,m,e,a) \\
&= \sum_{e,a} P(B)P(e)P(a|B,e)P(j|a)P(m|a) \\
&= \sum_{e} P(B)P(e)\sum_{a} P(a|B,e)P(j|a)P(m|a) \\
&= \sum_{e} P(B)P(e)f_1(B,e,j,m) \\
&= P(B)\sum_{e} P(e)f_1(B,e,j,m) \\
&= P(B)f_2(B,j,m)
\end{aligned}$$

marginal obtained from joint by summing out

use Bayes' net joint distribution expression

use x*(y+z) = xy + xz

joining on a, and then summing out gives f_1

use x*(y+z) = xy + xz

joining on e, and then summing out gives f_2

All we are doing is exploiting uwy + uwz + uxy + uxz + vwy + vwz + vxy +vxz = (u+v)(w+x)(y+z) to improve computational efficiency!

Important : Conditional Independence

- Every node is independent from its non-descendants given its parents
- Can we generalize this?

We will see the definition of D-separation in a minute

- *D-separation* generalizes this:

It is guaranteed that $X \perp\!\!\!\perp Y | Z$

if and only if

Every node in X is D-separated from every node in Y by Z

D-separation: Outline

- Study independence properties for triples
- Analyze complex cases in terms of member triples
- D-separation: a condition / algorithm for answering such queries

Causal Chains

- This configuration is a "causal chain"



X: Low pressure Y: Rain Z: Traffic

$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

- Guaranteed X independent of Z ? *No!*
  - One example set of CPTs for which X is not independent of Z is sufficient to show this independence is not guaranteed.
  - Example:
    - Low pressure causes rain causes traffic, high pressure causes no rain causes no traffic
    - In numbers:

      P(+y | +x) = 1, P(-y | - x) = 1,
      P(+z | +y) = 1, P(-z | -y) = 1

Causal Chains

- This configuration is a "causal chain"



X: Low pressure Y: Rain Z: Traffic

$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

- Guaranteed X independent of Z given Y?

$$P(z|x, y) = \frac{P(x, y, z)}{P(x, y)}$$

$$= \frac{P(x)P(y|x)P(z|y)}{P(x)P(y|x)}$$

$$= P(z|y)$$

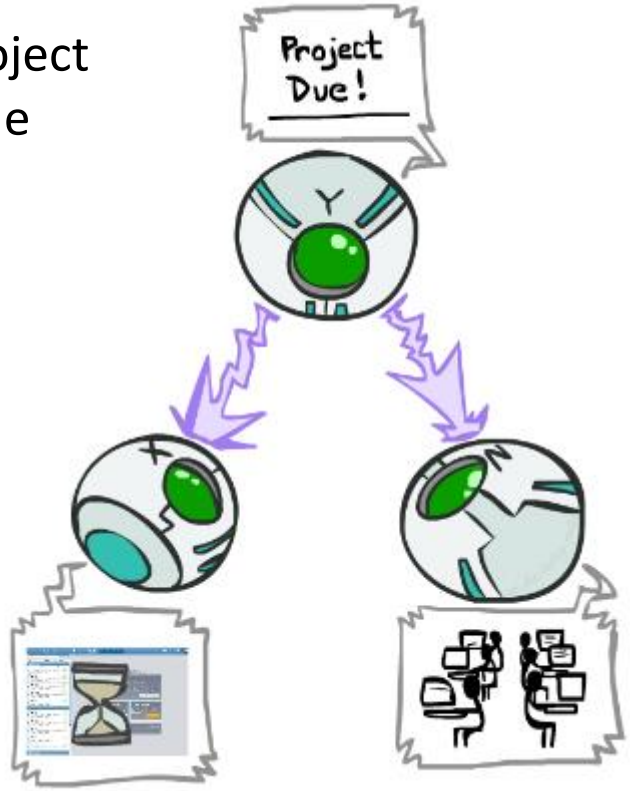
Yes!

- Evidence along the chain "blocks" the influence

Common Cause ("Confounder")

- This configuration is a "common cause"

Y: Project due



X: Forums busy

Z: Lab full

$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

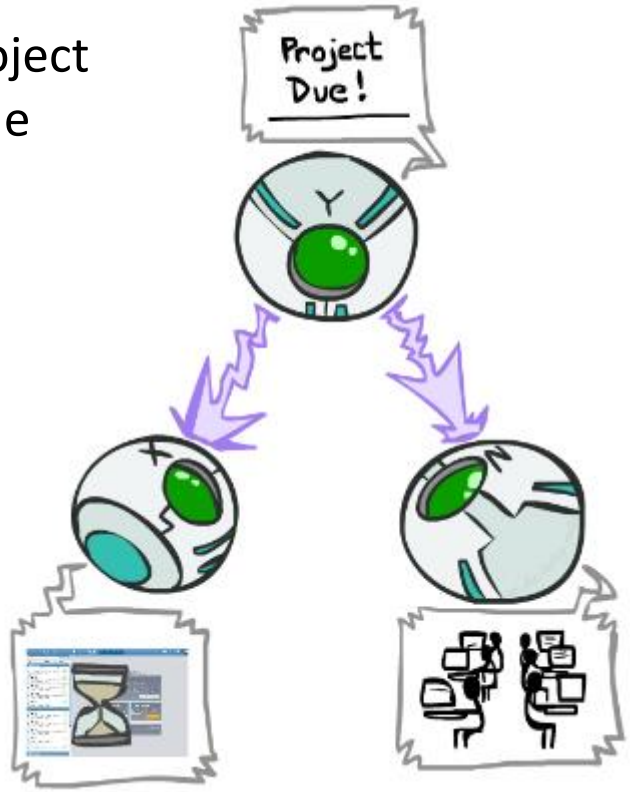
- Guaranteed X independent of Z ? *No!*
  - One example set of CPTs for which X is not independent of Z is sufficient to show this independence is not guaranteed.
  - Example:
    - Project due causes both forums busy and lab full
    - In numbers:

      P(+x | +y) = 1, P(-x | -y) = 1,
      P(+z | +y) = 1, P(-z | -y) = 1

Common Cause

- This configuration is a "common cause"

Y: Project due



X: Forums busy

Z: Lab full

$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

- Guaranteed X and Z independent given Y?

$$P(z|x, y) = \frac{P(x, y, z)}{P(x, y)}$$

$$= \frac{P(y)P(x|y)P(z|y)}{P(y)P(x|y)}$$

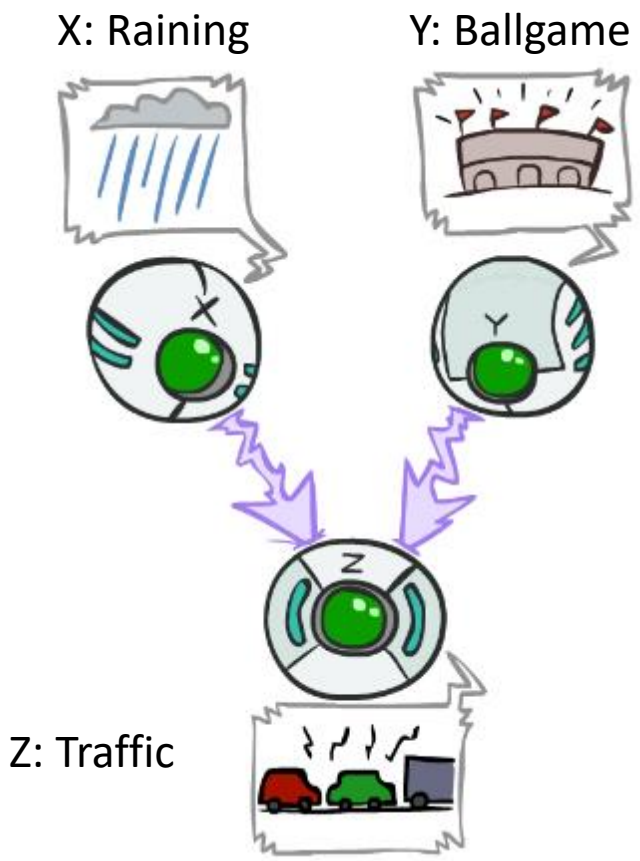
$$= P(z|y)$$

Yes!

- Observing the cause blocks influence between effects.

Common Effect ("Collider")

- Last configuration: two causes of one effect (v-structures)



- Are X and Y independent?
  - *Yes*: the ballgame and the rain cause traffic, but they are not correlated
  - Still need to prove they must be (try it!)
- Are X and Y independent given Z?
  - *No*: seeing traffic puts the rain and the ballgame in competition as explanation.
- This is backwards from the other cases
  - Observing an effect activates influence between possible causes.

The General Case

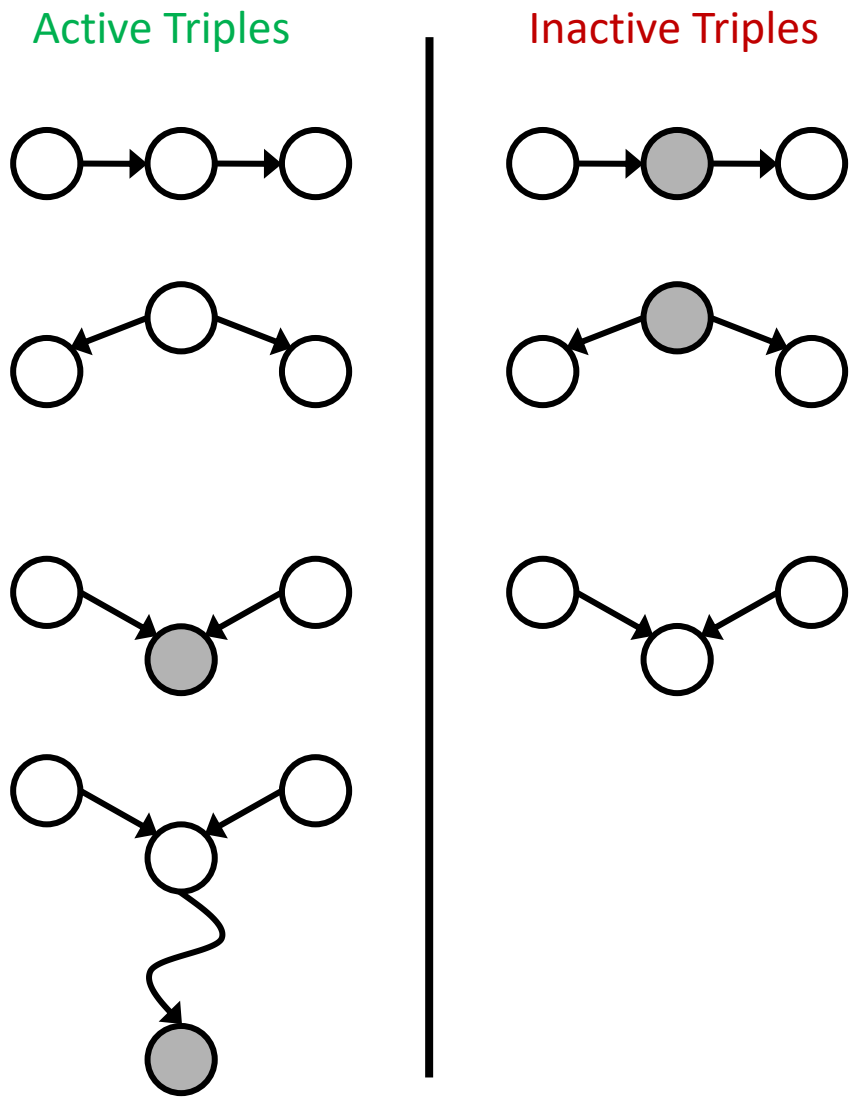


Active / Inactive Paths

- Question: Are X and Y conditionally independent given evidence variables {Z}?
  - Yes, if X and Y "d-separated" by Z
  - Consider all (undirected) paths from X to Y
  - No active paths = independence!

- A path is active if each triple is active:
  - Causal chain $A \rightarrow B \rightarrow C$ where B is unobserved (either direction)
  - Common cause $A \leftarrow B \rightarrow C$ where B is unobserved
  - Common effect (aka v-structure)
    $A \rightarrow B \leftarrow C$ where B *or one of its descendents* is observed

- All it takes to block a path is a single inactive segment



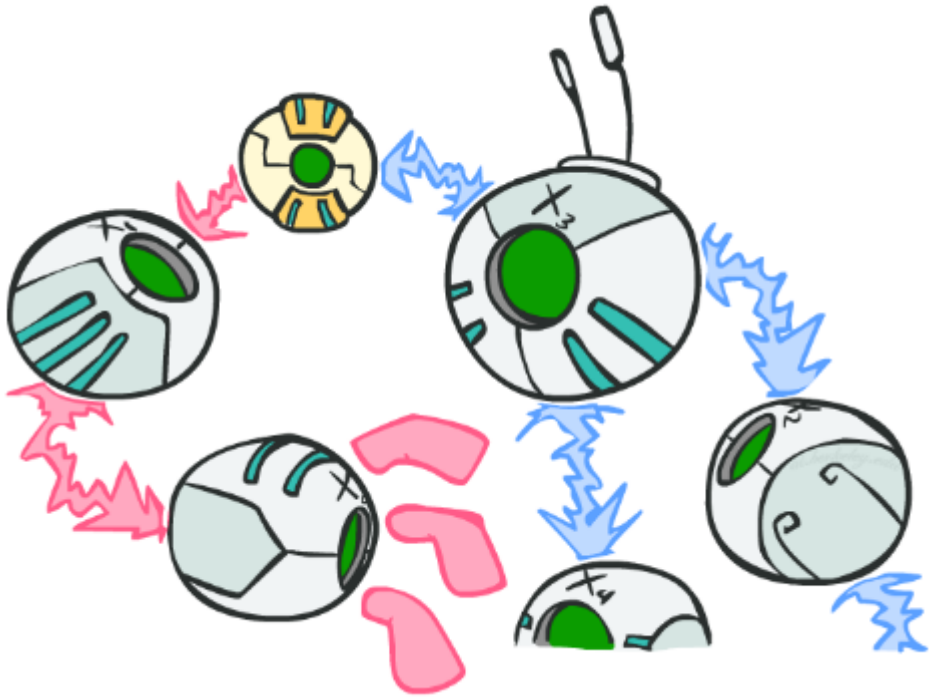
D-Separation

- Query: $X_i \perp\!\!\!\perp X_j | \{X_{k_1}, ..., X_{k_n}\}$?
- Check all (undirected!) paths between X_i and X_j
  - If one or more active, then independence not guaranteed

$$X_i \not\perp\!\!\!\perp X_j | \{X_{k_1}, ..., X_{k_n}\}$$

  - Otherwise (i.e. if all paths are inactive), then independence is guaranteed

$$X_i \perp\!\!\!\perp X_j | \{X_{k_1}, ..., X_{k_n}\}$$

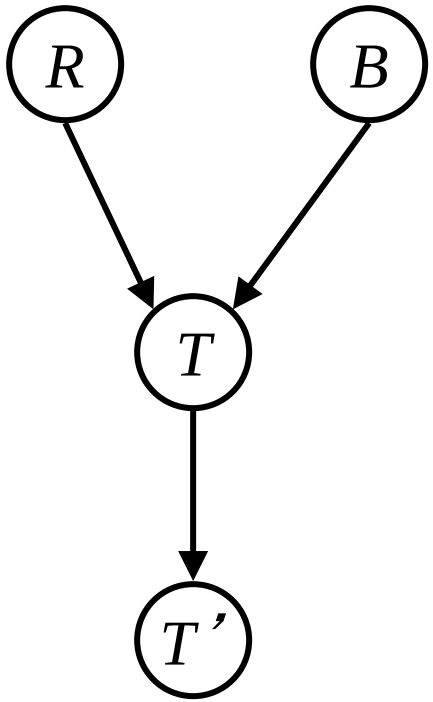


Example

$R \perp\!\!\!\perp B$ *Yes*

$R \perp\!\!\!\perp B | T$

$R \perp\!\!\!\perp B | T'$



Example

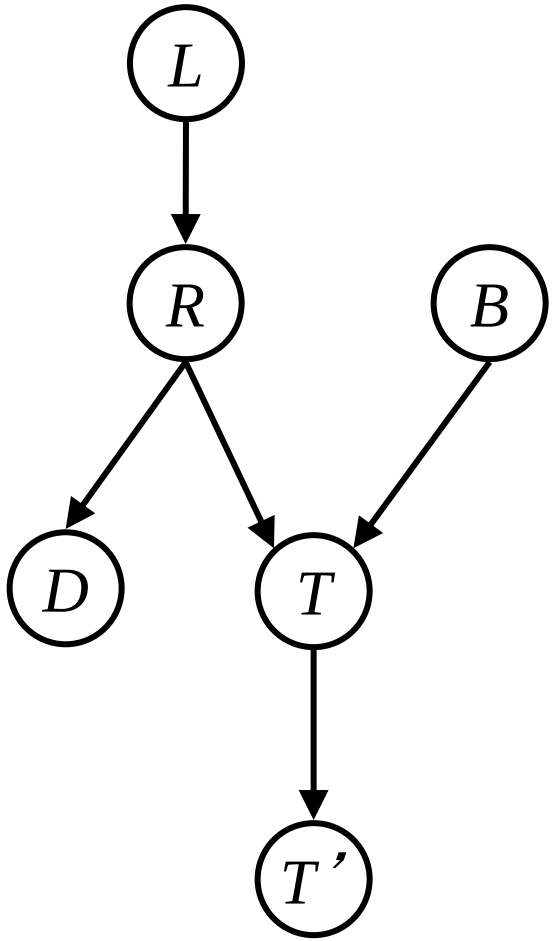
$L \perp\!\!\!\perp T' | T$ *Yes*

$L \perp\!\!\!\perp B$ *Yes*

$L \perp\!\!\!\perp B | T$

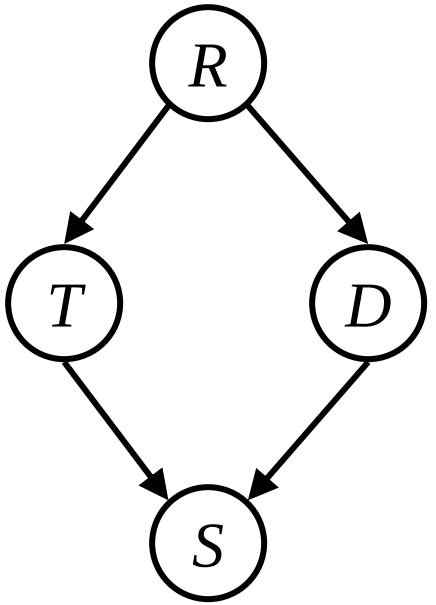
$L \perp\!\!\!\perp B | T'$

$L \perp\!\!\!\perp B | T, R$ *Yes*



Example

- Variables:
  - R: Raining
  - T: Traffic
  - D: Roof drips
  - S: I'm sad
- Questions:



$$T \perp\!\!\!\perp D$$

$$T \perp\!\!\!\perp D | R \qquad \textit{Yes}$$

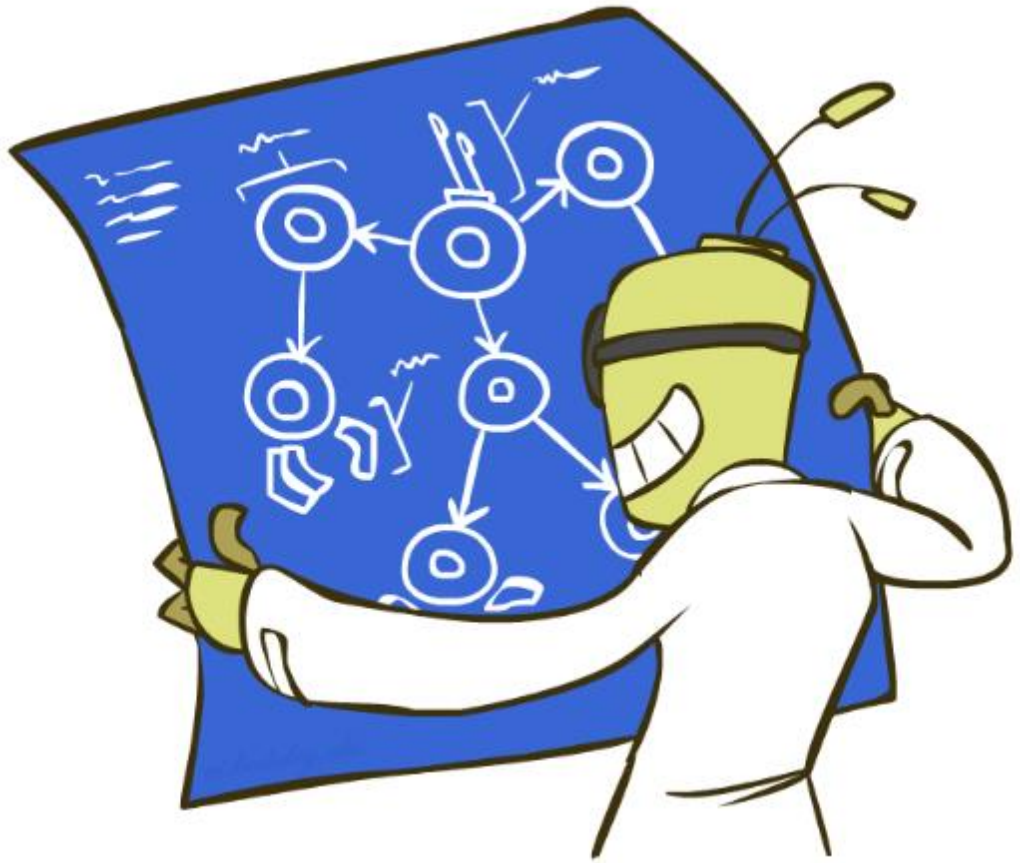
$$T \perp\!\!\!\perp D | R, S$$

Structure Implications

- Given a Bayes net structure, can run d-separation algorithm to build a complete list of conditional independences that are necessarily true of the form

$$X_i \perp\!\!\!\perp X_j | \{X_{k_1}, ..., X_{k_n}\}$$

- This list determines the set of probability distributions that can be represented

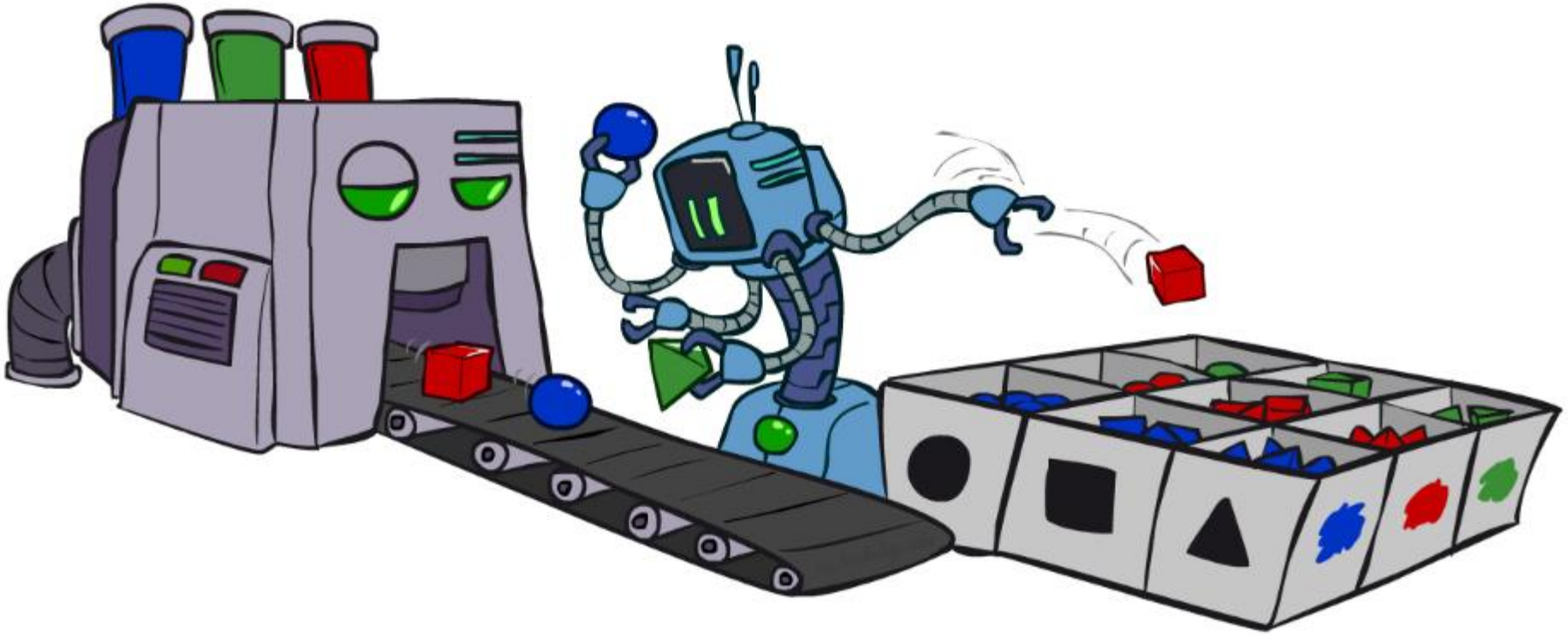


Bayes Nets Representation Summary

- Bayes nets compactly encode joint distributions
- Guaranteed independencies of distributions can be deduced from BN graph structure
- D-separation gives precise conditional independence guarantees from graph alone
- A Bayes' net's joint distribution may have further (conditional) independence that is not detectable until you inspect its specific distribution

CS 188: Artificial Intelligence

Bayes' Nets: Sampling



Instructors: Dan Klein and Pieter Abbeel --- University of California, Berkeley

[These slides were created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All CS188 materials are available at http://ai.berkeley.edu.]

Worst Case Complexity of Inference?

- CSP:

$$(x_1 \vee x_2 \vee \neg x_3) \wedge (\neg x_1 \vee x_3 \vee \neg x_4) \wedge (x_2 \vee \neg x_2 \vee x_4) \wedge (\neg x_3 \vee \neg x_4 \vee \neg x_5) \wedge (x_2 \vee x_5 \vee x_7) \wedge (x_4 \vee x_5 \vee x_6) \wedge (\neg x_5 \vee x_6 \vee \neg x_7) \wedge (\neg x_5 \vee \neg x_6 \vee x_7)$$

$$P(X_i = 0) = P(X_i = 1) = 0.5$$

$$Y_1 = X_1 \vee X_2 \vee \neg X_3$$

...

$$Y_8 = \neg X_5 \vee X_6 \vee X_7$$

$$Y_{1,2} = Y_1 \wedge Y_2$$

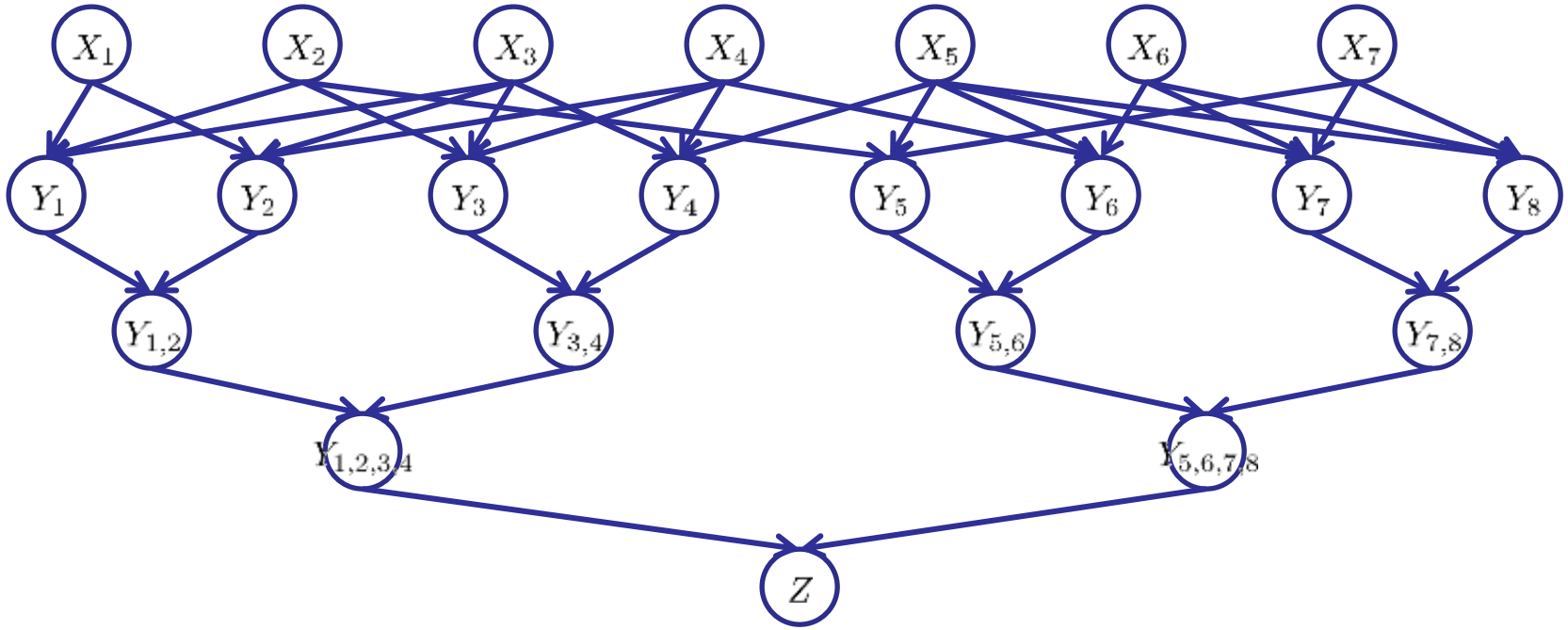
...

$$Y_{7,8} = Y_7 \wedge Y_8$$

$$Y_{1,2,3,4} = Y_{1,2} \wedge Y_{3,4}$$

$$Y_{5,6,7,8} = Y_{5,6} \wedge Y_{7,8}$$

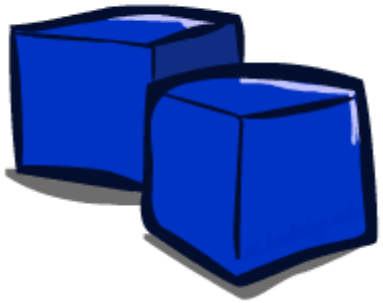
$$Z = Y_{1,2,3,4} \wedge Y_{5,6,7,8}$$



- If we can answer P(z) equal to zero or not, we answered whether the 3-SAT problem has a solution.
- Hence inference in Bayes' nets is NP-hard. No known efficient probabilistic inference in general.

Approximate Inference: Sampling

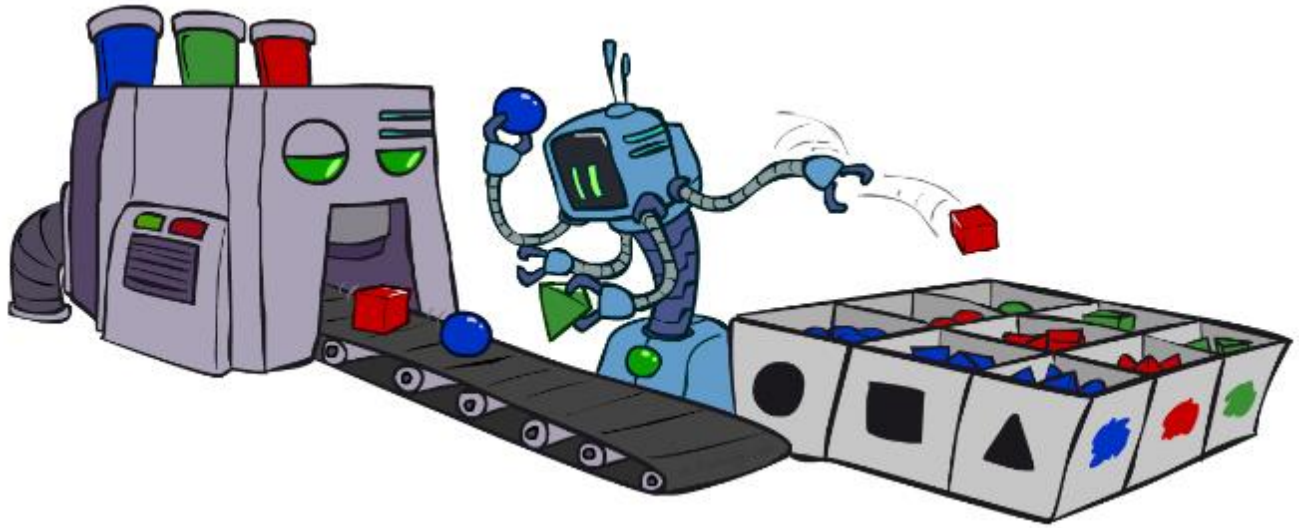






Sampling

- Sampling is a lot like repeated simulation
  - Predicting the weather, basketball games, …
- Basic idea
  - Draw N samples from a sampling distribution S
  - Compute an approximate posterior probability
  - Show this converges to the true probability P
- Why sample?
  - Learning: get samples from a distribution you don't know
  - Inference: getting a sample is faster than computing the right answer (e.g. with variable elimination)



Sampling

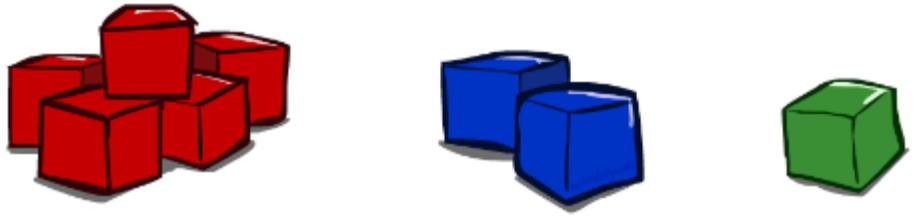
- Sampling from given distribution
  - Step 1: Get sample *u* from uniform distribution over [0, 1)
    - E.g. random() in python
  - Step 2: Convert this sample *u* into an outcome for the given distribution by having each target outcome associated with a sub-interval of [0,1) with sub-interval size equal to probability of the outcome

- Example

C	P(C)
red	0.6
green	0.1
blue	0.3

$$0 \leq u < 0.6, \rightarrow C = red$$
$$0.6 \leq u < 0.7, \rightarrow C = green$$
$$0.7 \leq u < 1, \rightarrow C = blue$$

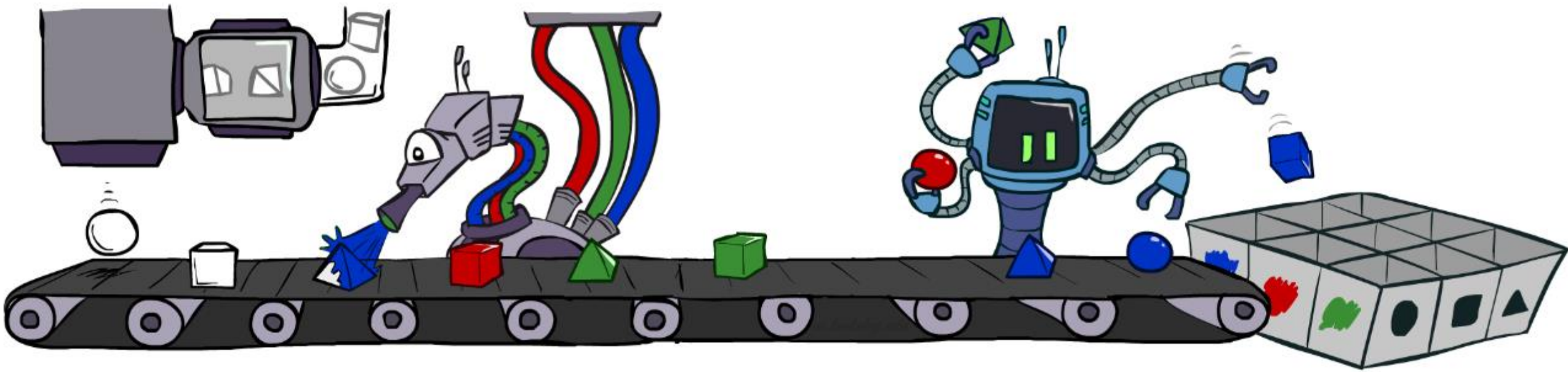
  - If random() returns $u = 0.83$, then our sample is C = blue
  - E.g, after sampling 8 times:



Sampling in Bayes' Nets

- Prior Sampling
- Rejection Sampling
- Likelihood Weighting
- Gibbs Sampling

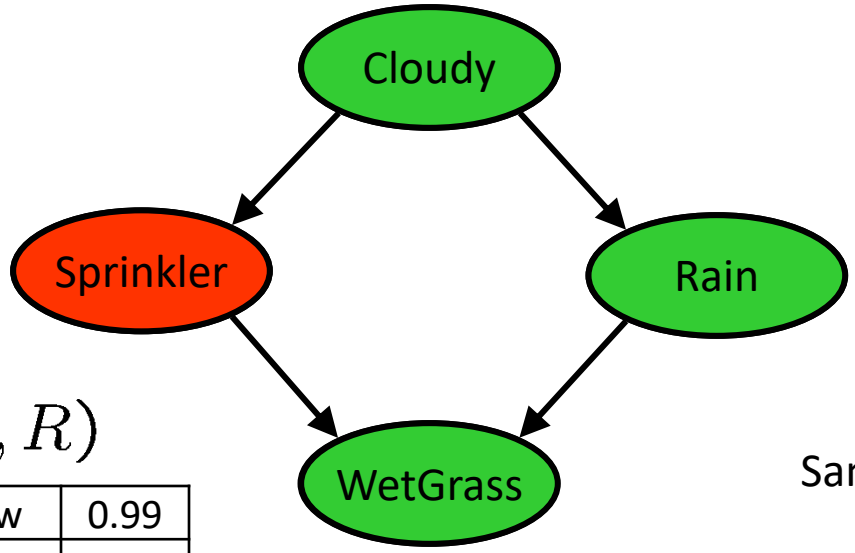
Prior Sampling



Prior Sampling

$P(C)$

+c	0.5
-c	0.5



$P(S|C)$

+c	+s	0.1
	-s	0.9
-c	+s	0.5
	-s	0.5

$P(R|C)$

+c	+r	0.8
	-r	0.2
-c	+r	0.2
	-r	0.8

$P(W|S,R)$

+s	+r	+w	0.99
		-w	0.01
	-r	+w	0.90
		-w	0.10
-s	+r	+w	0.90
		-w	0.10
	-r	+w	0.01
		-w	0.99

Samples:

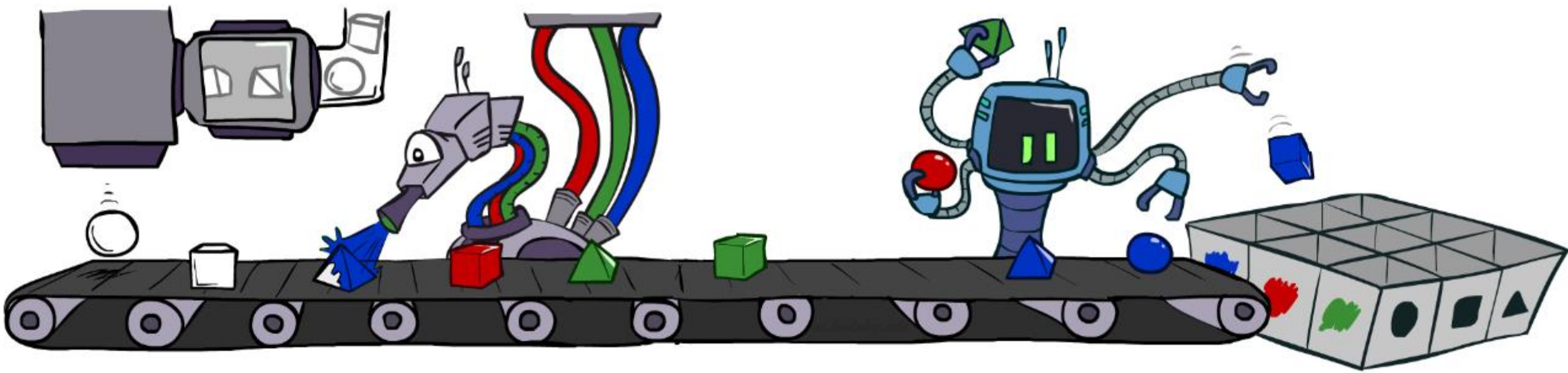
+c, -s, +r, +w

-c, +s, -r, +w

…

Prior Sampling

- For i = 1, 2, …, n
  - Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
- Return $(x_1, x_2, \ldots, x_n)$



Prior Sampling

- This process generates samples with probability:

$$S_{PS}(x_1 \ldots x_n) = \prod_{i=1}^{n} P(x_i|\text{Parents}(X_i)) = P(x_1 \ldots x_n)$$

  …i.e. the BN's joint probability

- Let the number of samples of an event be $N_{PS}(x_1 \ldots x_n)$

- Then $$\begin{aligned} \lim_{N\to\infty} \hat{P}(x_1, \ldots, x_n) &= \lim_{N\to\infty} N_{PS}(x_1, \ldots, x_n)/N \\ &= S_{PS}(x_1, \ldots, x_n) \\ &= P(x_1 \ldots x_n) \end{aligned}$$

- I.e., the sampling procedure is consistent

Example

- We'll get a bunch of samples from the BN:

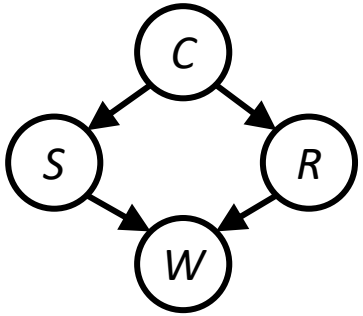
  +c, -s, +r, +w

  +c, +s, +r, +w

  -c, +s, +r, -w

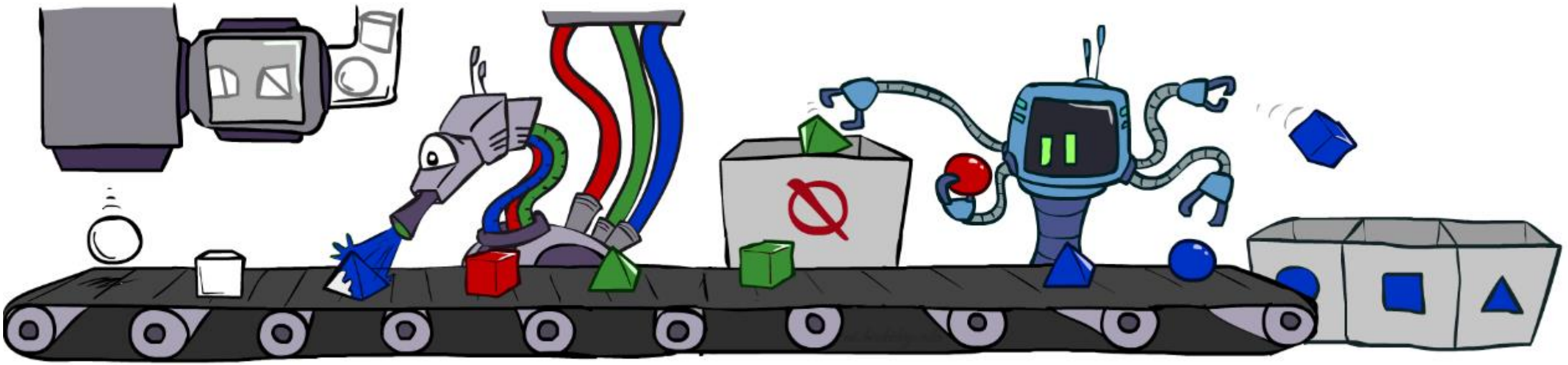
  +c, -s, +r, +w

  -c, -s, -r, +w



- If we want to know P(W)
  - We have counts <+w:4, -w:1>
  - Normalize to get P(W) = <+w:0.8, -w:0.2>
  - This will get closer to the true distribution with more samples
  - Can estimate anything else, too
  - What about P(C | +w)? P(C | +r, +w)? P(C | -r, -w)?
  - Fast: can use fewer samples if less time (what's the drawback?)

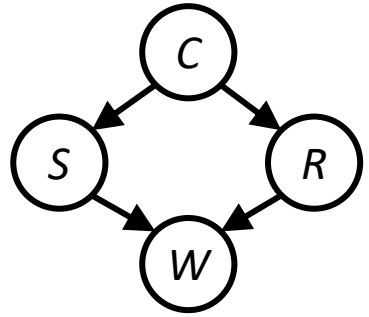
Rejection Sampling



Rejection Sampling

- Let's say we want P(C)
  - No point keeping all samples around
  - Just tally counts of C as we go

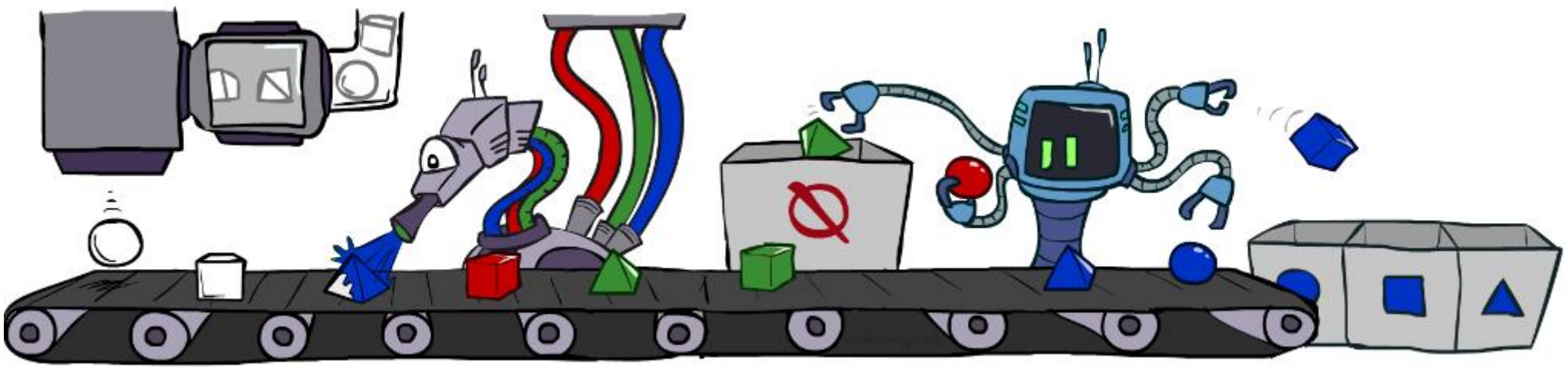
- Let's say we want P(C | +s)
  - Same thing: tally C outcomes, but ignore (reject) samples which don't have S=+s
  - This is called rejection sampling
  - It is also consistent for conditional probabilities (i.e., correct in the limit)



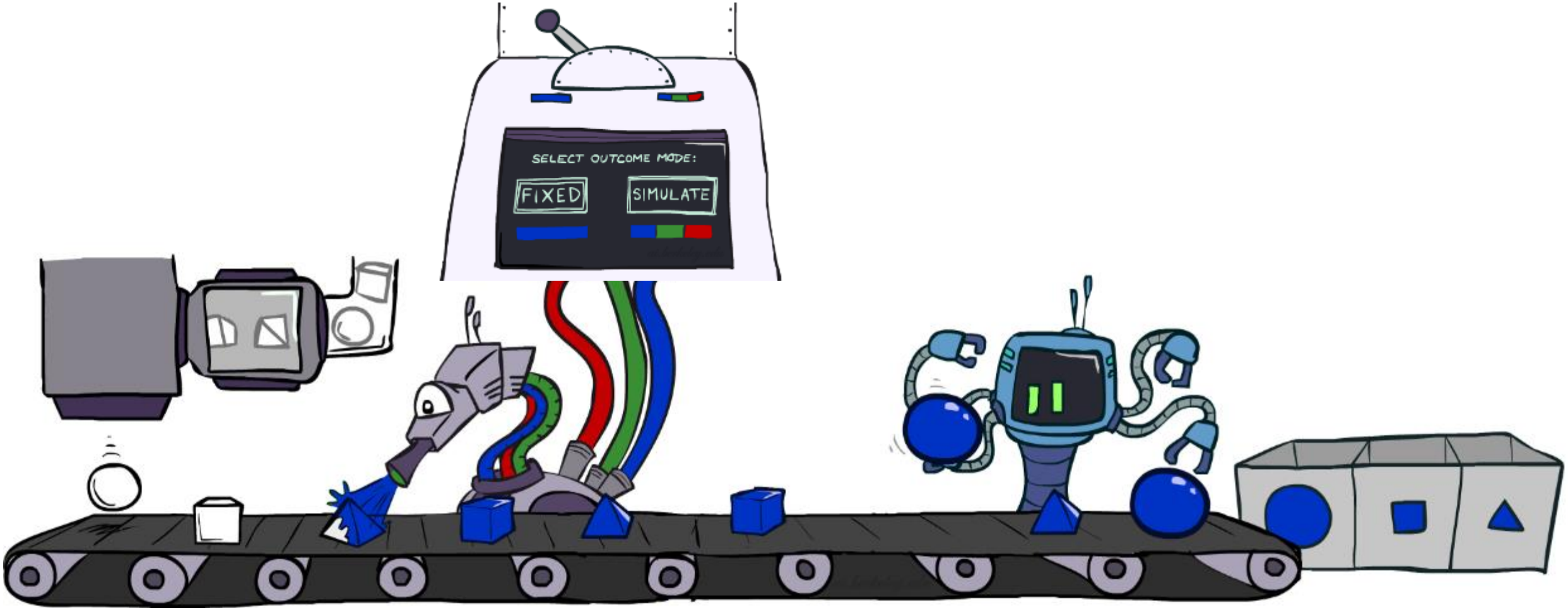
+c, -s, +r, +w
+c, +s, +r, +w
-c, +s, +r, -w
+c, -s, +r, +w
-c, -s, -r, +w

Rejection Sampling

- Input: evidence instantiation
- For i = 1, 2, …, n
  - Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
  - If x_i not consistent with evidence
    - Reject: return – no sample is generated in this cycle
- Return $(x_1, x_2, \ldots, x_n)$

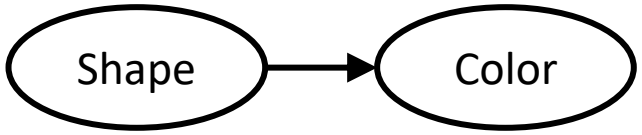


Likelihood Weighting

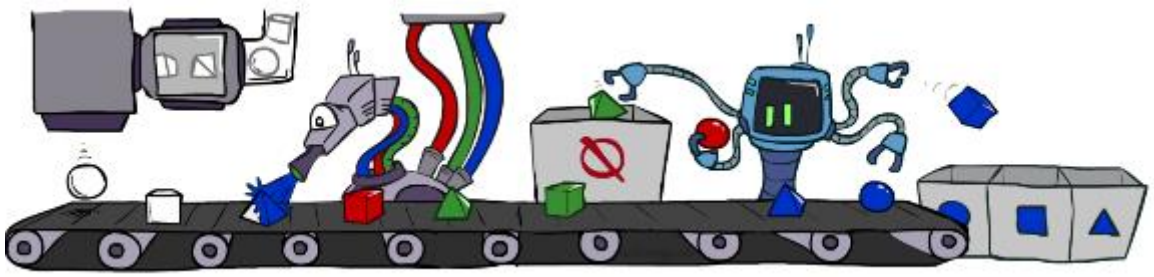


Likelihood Weighting

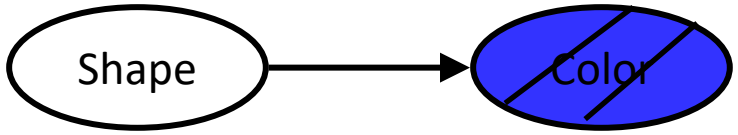
- Problem with rejection sampling:
  - If evidence is unlikely, rejects lots of samples
  - Evidence not exploited as you sample
  - Consider P(Shape | blue)



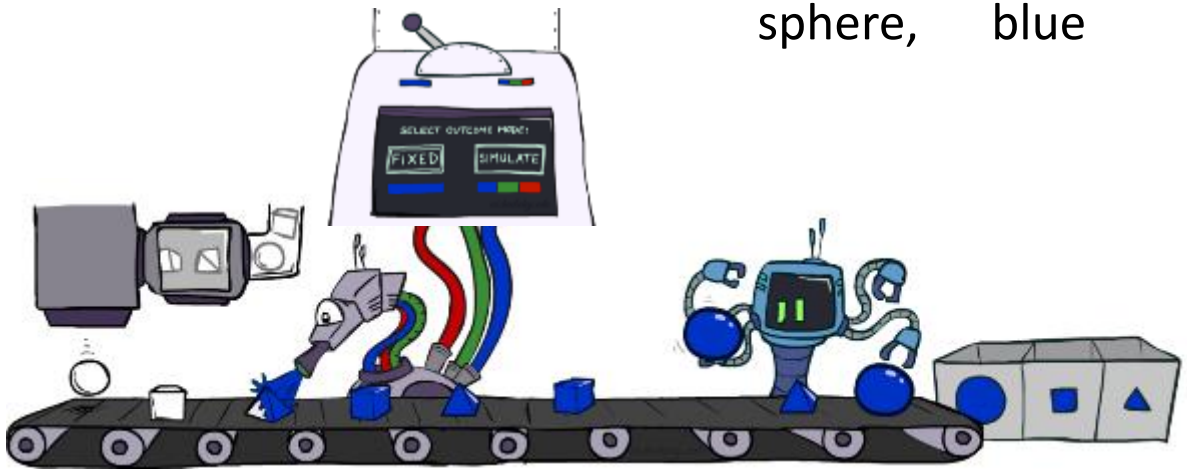
~~pyramid, green~~
~~pyramid, red~~
sphere, blue
~~cube, red~~
~~sphere, green~~



- Idea: fix evidence variables and sample the rest
  - Problem: sample distribution not consistent!
  - Solution: weight by probability of evidence given parents



pyramid, blue
pyramid, blue
sphere, blue
cube, blue
sphere, blue



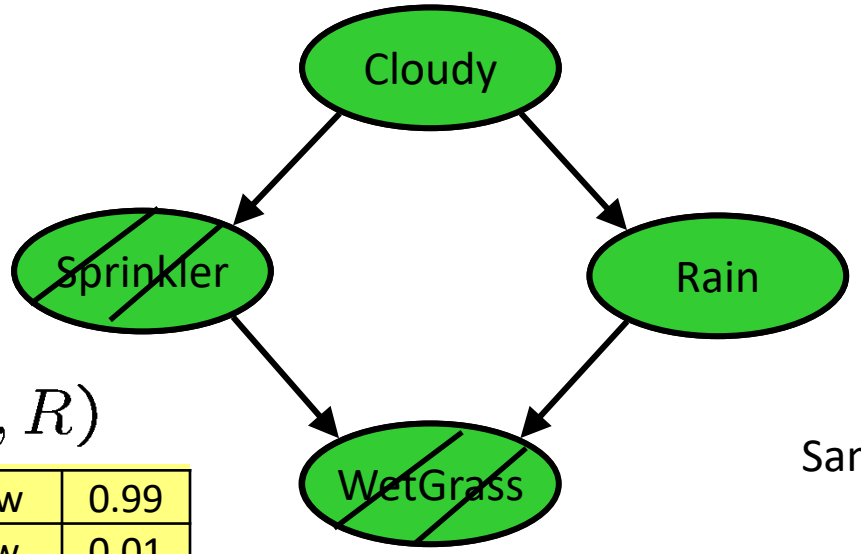
Likelihood Weighting

$P(C)$

+c	0.5
-c	0.5

$P(S|C)$

+c	+s	0.1
	-s	0.9
-c	+s	0.5
	-s	0.5



$P(R|C)$

+c	+r	0.8
	-r	0.2
-c	+r	0.2
	-r	0.8

$P(W|S,R)$

+s	+r	+w	0.99
		-w	0.01
	-r	+w	0.90
		-w	0.10
-s	+r	+w	0.90
		-w	0.10
	-r	+w	0.01
		-w	0.99

Samples:

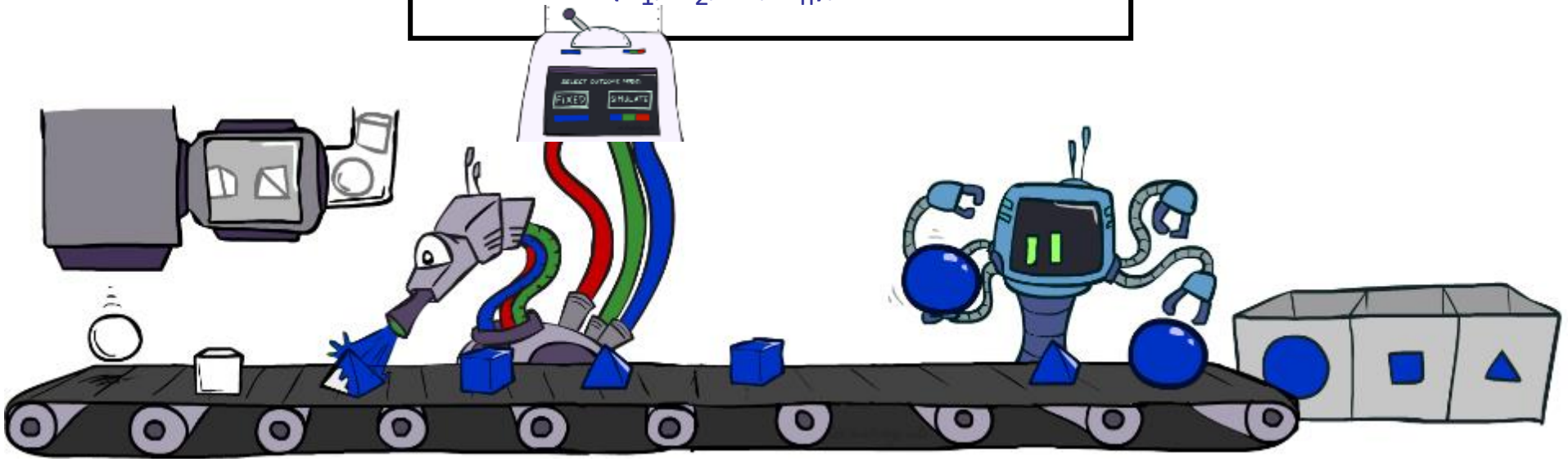
+c, +s, +r, +w

…

$$w = 1.0 \times 0.1 \times 0.99$$

Likelihood Weighting

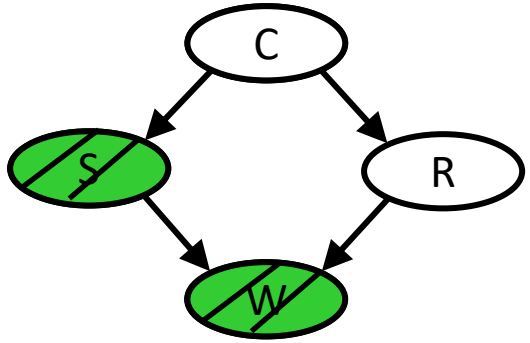
- Input: evidence instantiation
- w = 1.0
- for i = 1, 2, …, n
  - if X_i is an evidence variable
    - X_i = observation x_i for X_i
    - Set w = w * $P(x_i \mid Parents(X_i))$
  - else
    - Sample x_i from $P(X_i \mid Parents(X_i))$
- return $(x_1, x_2, \ldots, x_n)$, w



Likelihood Weighting

- Sampling distribution if z sampled and e fixed evidence

$$S_{WS}(\mathbf{z}, \mathbf{e}) = \prod_{i=1}^{l} P(z_i | \text{Parents}(Z_i))$$



- Now, samples have weights

$$w(\mathbf{z}, \mathbf{e}) = \prod_{i=1}^{m} P(e_i | \text{Parents}(E_i))$$

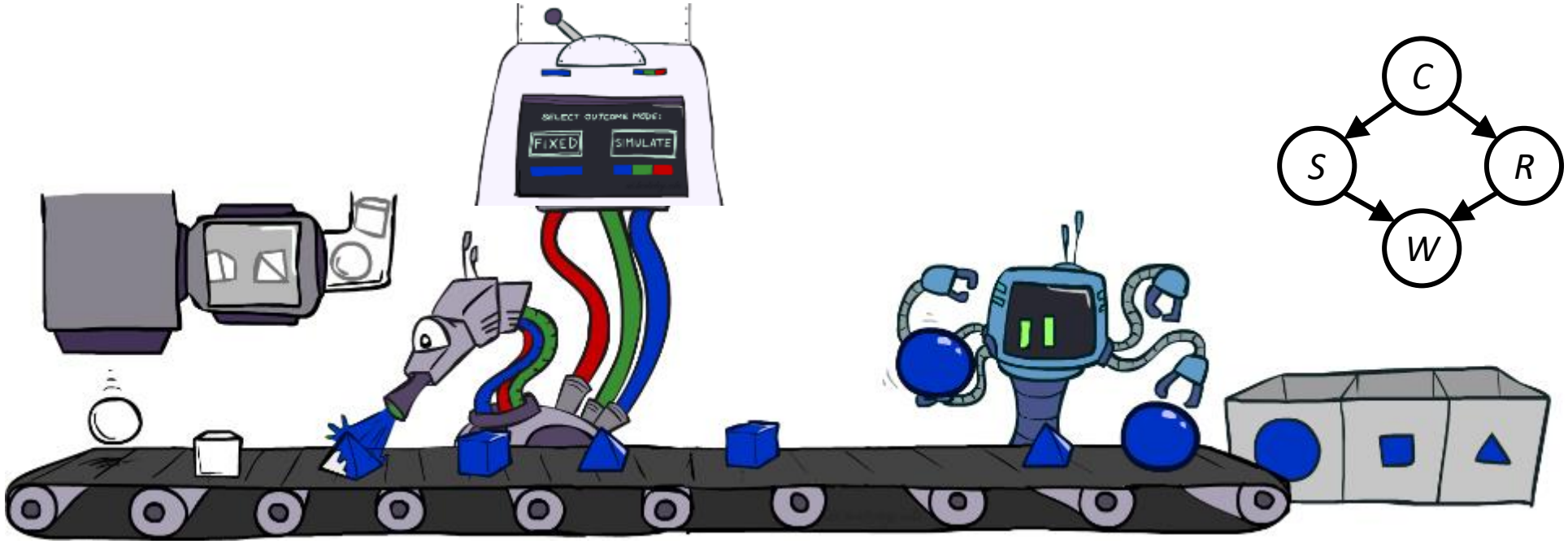
- Together, weighted sampling distribution is consistent

$$S_{\text{WS}}(z, e) \cdot w(z, e) = \prod_{i=1}^{l} P(z_i | \text{Parents}(z_i)) \prod_{i=1}^{m} P(e_i | \text{Parents}(e_i))$$

$$= P(\mathbf{z}, \mathbf{e})$$

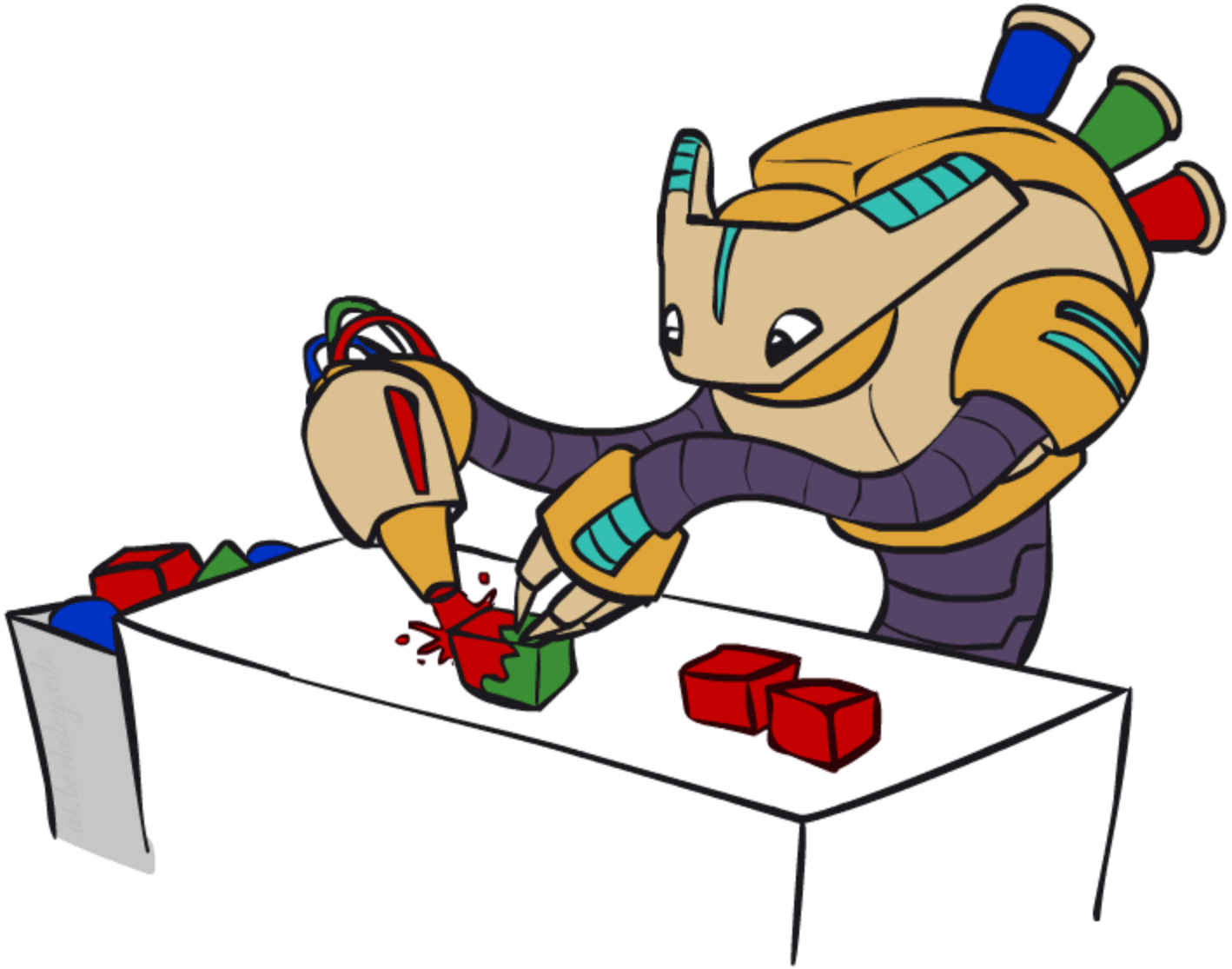
Likelihood Weighting

- Likelihood weighting is good
  - We have taken evidence into account as we generate the sample
  - E.g. here, W's value will get picked based on the evidence values of S, R
  - More of our samples will reflect the state of the world suggested by the evidence



- Likelihood weighting doesn't solve all our problems
  - Evidence influences the choice of downstream variables, but not upstream ones (C isn't more likely to get a value matching the evidence)
- We would like to consider evidence when we sample every variable (leads to Gibbs sampling)

Gibbs Sampling

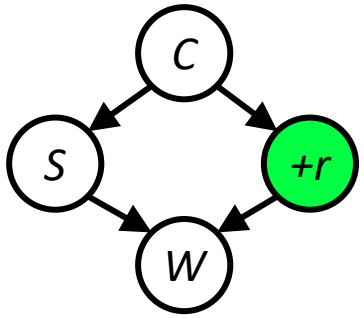


Gibbs Sampling

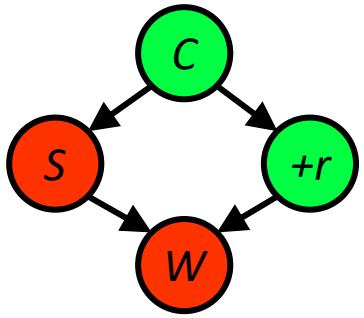
- *Procedure:* keep track of a full instantiation $x_1, x_2, \ldots, x_n$. Start with an arbitrary instantiation consistent with the evidence. Sample one variable at a time, conditioned on all the rest, but keep evidence fixed. Keep repeating this for a long time.
- *Property:* in the limit of repeating this infinitely many times the resulting samples come from the correct distribution (i.e. conditioned on evidence).
- *Rationale*: both upstream and downstream variables condition on evidence.
- In contrast: likelihood weighting only conditions on upstream evidence, and hence weights obtained in likelihood weighting can sometimes be very small. Sum of weights over all samples is indicative of how many "effective" samples were obtained, so we want high weight.

Gibbs Sampling Example: P(S | +r)

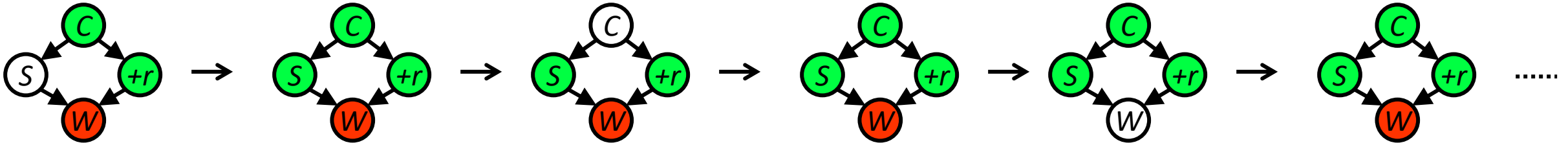
- Step 1: Fix evidence
  - R = +r



- Step 2: Initialize other variables
  - Randomly



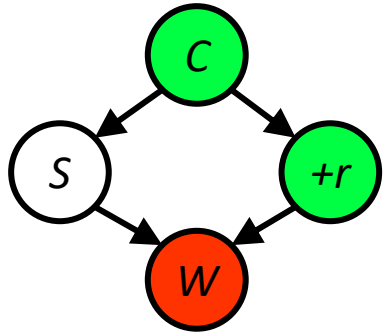
- Steps 3: Repeat
  - Choose a non-evidence variable X
  - Resample X from P(X | all other variables)



Sample from $P(S|+c,-w,+r)$ Sample from $P(C|+s,-w,+r)$ Sample from $P(W|+s,+c,+r)$

Efficient Resampling of One Variable

- Sample from P(S | +c, +r, -w)

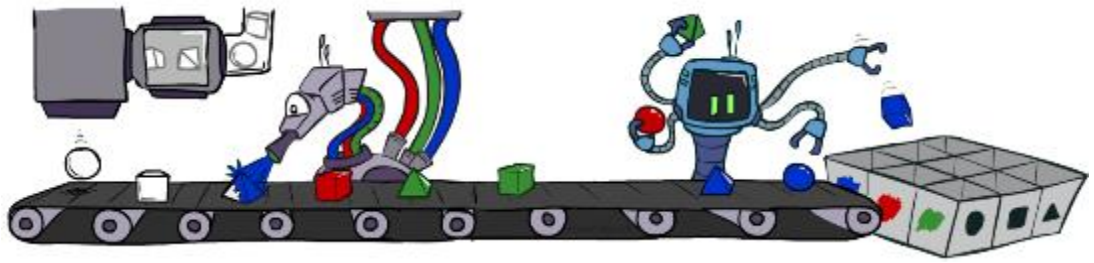


$$
\begin{aligned}
P(S|+c,+r,-w) &= \frac{P(S,+c,+r,-w)}{P(+c,+r,-w)} \\
&= \frac{P(S,+c,+r,-w)}{\sum_s P(s,+c,+r,-w)} \\
&= \frac{P(+c)P(S|+c)P(+r|+c)P(-w|S,+r)}{\sum_s P(+c)P(s|+c)P(+r|+c)P(-w|s,+r)} \\
&= \frac{P(+c)P(S|+c)P(+r|+c)P(-w|S,+r)}{P(+c)P(+r|+c)\sum_s P(s|+c)P(-w|s,+r)} \\
&= \frac{P(S|+c)P(-w|S,+r)}{\sum_s P(s|+c)P(-w|s,+r)}
\end{aligned}
$$

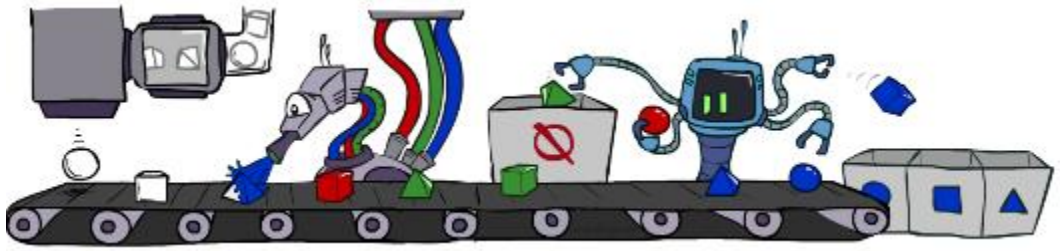
- Many things cancel out – only CPTs with S remain!
- More generally: only CPTs that have resampled variable need to be considered, and joined together

Bayes' Net Sampling Summary

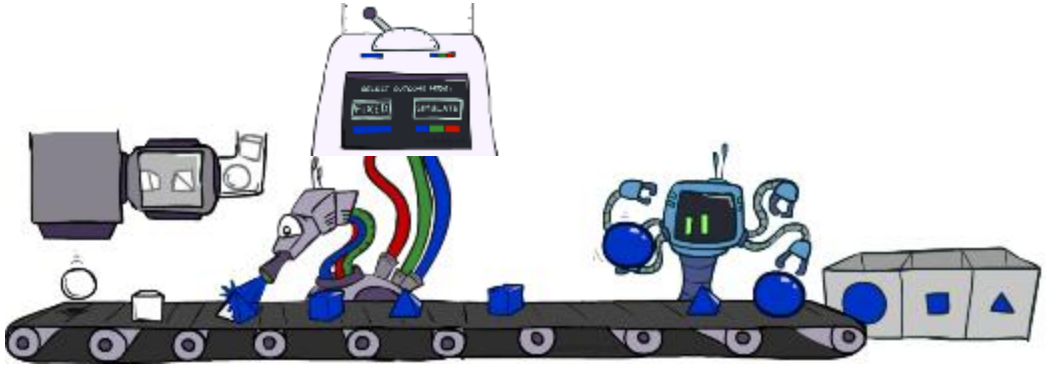
- Prior Sampling P(Q)



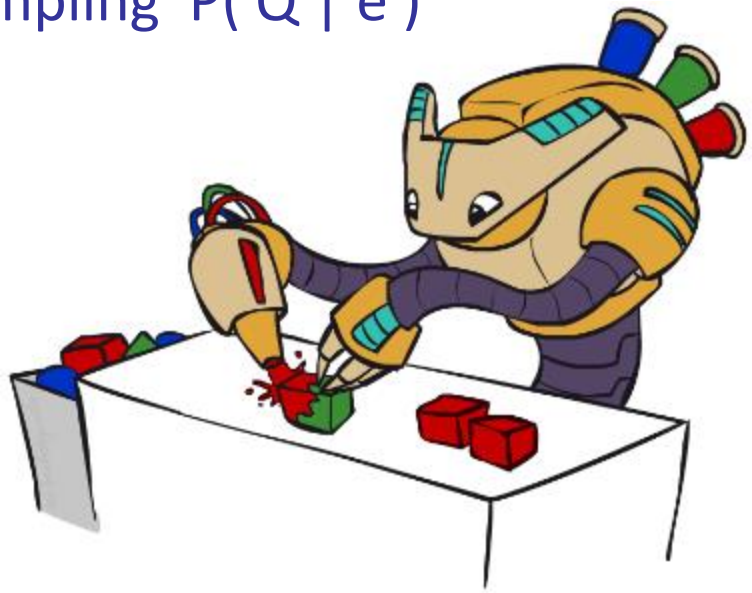
- Rejection Sampling P(Q | e)



- Likelihood Weighting P(Q | e)



- Gibbs Sampling P(Q | e)

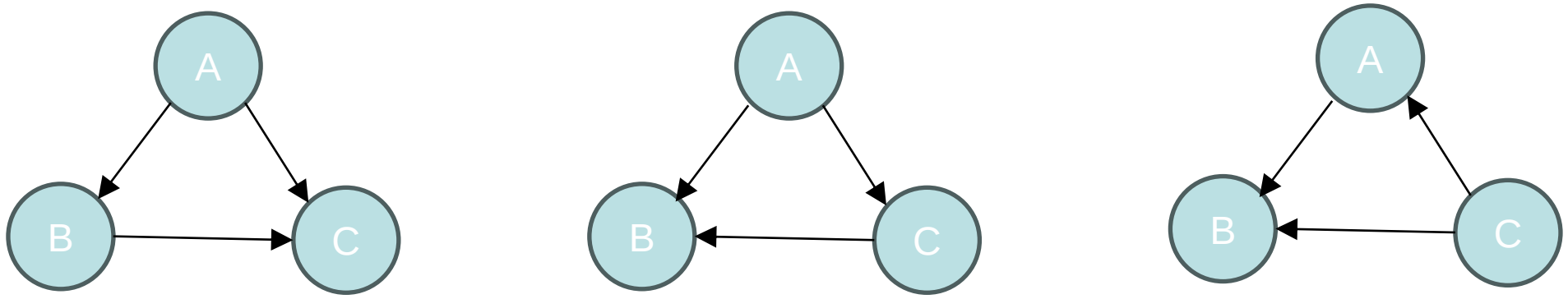


Further Reading on Gibbs Sampling*

- Gibbs sampling produces sample from the query distribution P(Q | e) in limit of re-sampling infinitely often
- Gibbs sampling is a special case of more general methods called Markov Chain Monte Carlo (MCMC) methods
  - Metropolis-Hastings is one of the more famous MCMC methods (in fact, Gibbs sampling is a special case of Metropolis-Hastings)
- You may read about Monte Carlo methods – they're just sampling

Note: Bayesian Nets and Causality

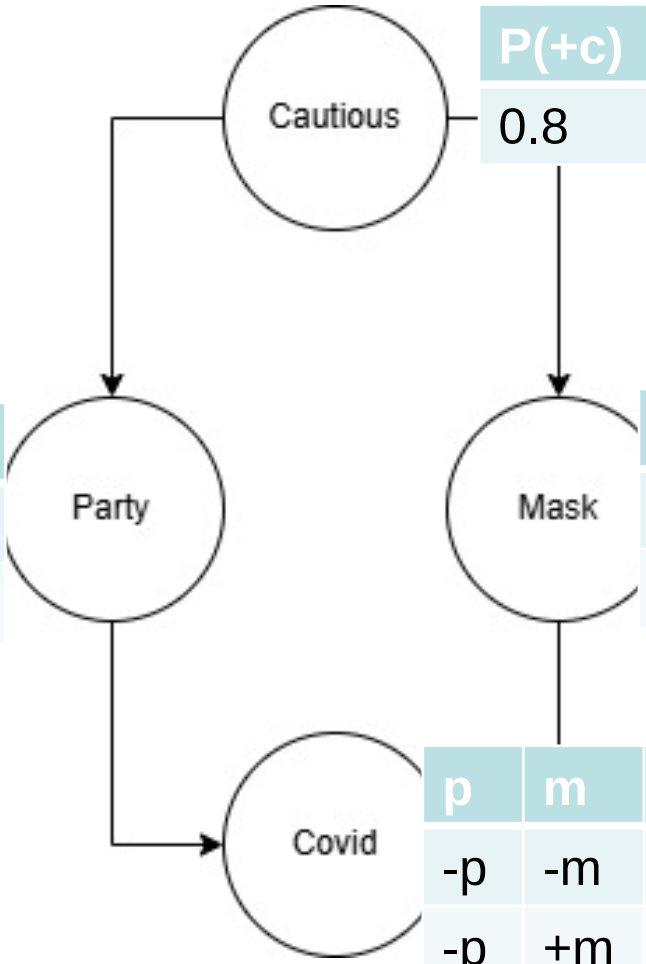
- Links in a Bayesian net are not necessarily causal



- All three Bayesian nets are equivalent
  - Dependencies indicate correlations, not causation
- Even if edges are causal, P(A|B) does not equal effect of B on A

Motivating Example: Causality

- What is P(Cov|M) ?
  - P(Cov|m+) around 12%
  - P(Cov|-m) around 72%
- So, effect of +m around 60% ?
- **Caveat! P(Cov|M) also measures correlation that is not causal!**



P(+c)
0.8

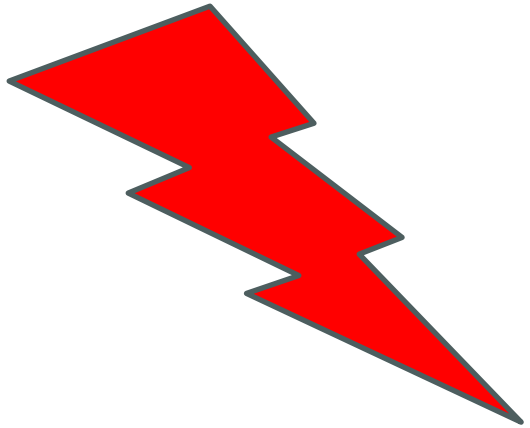
c	P(+p|c)
+c	0.05
-c	0.3

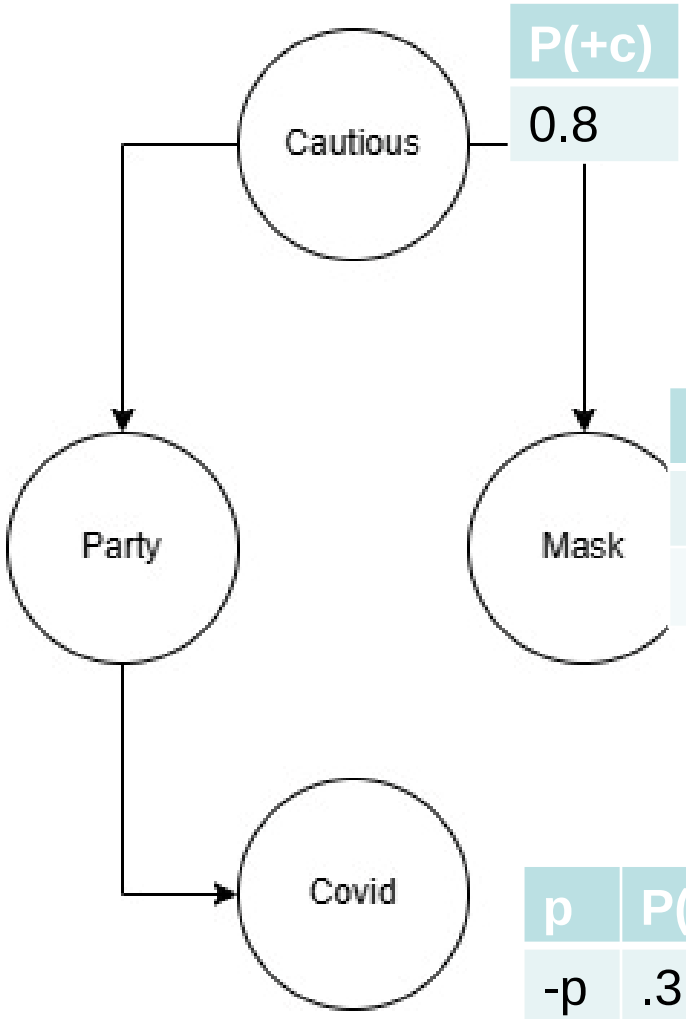
c	P(+m|c)
+c	0.05
-c	0.3

p	m	P(+cov|p,m)
-p	-m	.7
-p	+m	.1
+p	-m	.8
+p	+m	.4

Motivating Example: Causality

- What is P(Cov|M) ?
  - P(Cov|m+) around 34%
  - P(Cov|-m) around 42%
- So, effect of +m around 8% ?





P(+c)
0.8

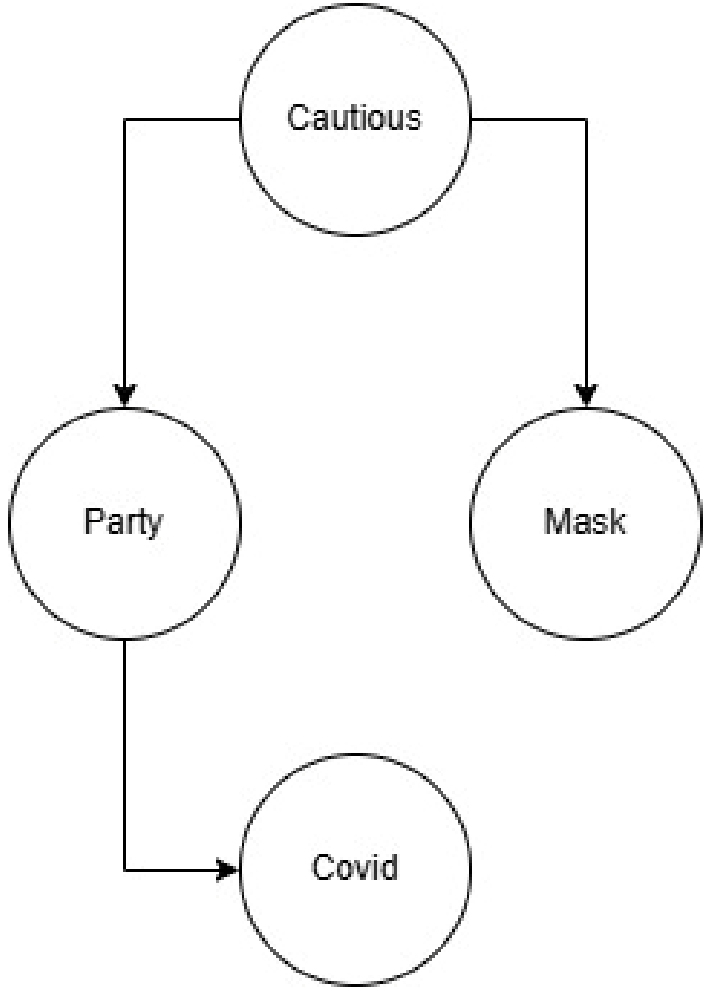
c	P(+m|c)
+c	0.05
-c	0.3

c	P(+p|c)
+c	0.05
-c	0.3

p	P(+cov|p)
-p	.3
+p	.8

Motivating Example: Causality

- If someone wears a mask, it is more likely that the person is cautious
- If a person is more cautious they are less likely they go party
- If a person is less likely to party, they are less likely to contract covid
- So, Mask does give info on Covid!



Causality

- Causal net = Bayesian net in which the links are *causal*
  - Mathematically formalized with *structural equations*
  - A parent of B: value of B is a function of A (+randomness)
- Intervention = changing a value of a variable
  - Intervention only affects descendants
  - E.g.: Smoking causes cancer
    - Intervene on smoking (e.g. stop smoking) affects presence of cancer
    - Intervene on cancer (e.g. surgery) does not affect smoking
- Causal nets allow to estimate *treatment effects*

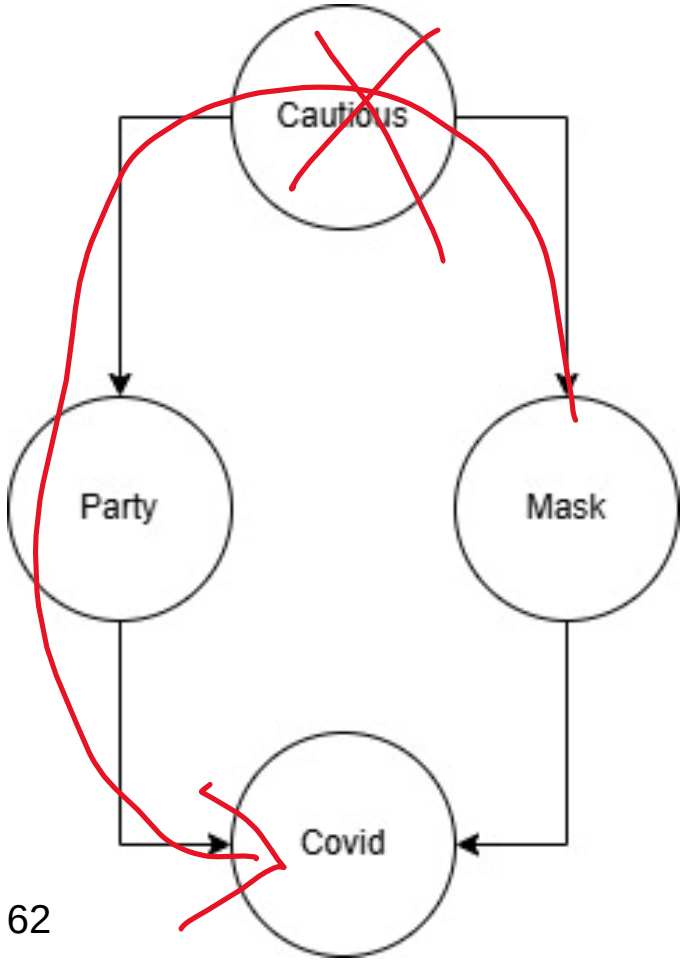
Backdoor Adjustment

- We want to estimate the effect of *taking* the action, not *observing* the action
  - What if a cautious person puts on a mask?
  - What if a non-cautious person does?
  - Average

$$P(\text{Cov} \mid \text{do}(M)) = \sum_c P(c)P(\text{Cov} \mid M, c)$$

Average Treatment Effect turns out to be 57%

- Called "backdoor adjustment"



Backdoor Adjustment

- We want to estimate the effect of *taking* the action, not *observing* the action
  - What if a party person puts on a mask?
  - What if a non-party person does?
  - Average

$$P(\text{Cov} \mid \text{do}(M))$$
$$= \sum_p P(p)P(\text{Cov} \mid M, p)$$

Average Treatment Effect turns out to be 57%

- Called "backdoor adjustment"

